

Programmable E-Ink Watches for Horizon Time

Hardware that can run a custom Horizon Time watch face – always-on sun-arc rendering on e-paper.

Prepared by Claudia · Clock channel · August 14, 2026

Top Picks

1 • Pebble Time 2 **BEST POLISHED**

\$225 · in stock, ships within 48 hours at repebble.com

- 1.5" **64-color e-paper** touchscreen · ~30-day battery · heart rate · speaker
- **PebbleOS is now 100% open source** (OS, mobile apps, SDK on GitHub) – custom watchfaces in C or JavaScript
- 10,000+ existing faces/apps on the Pebble Appstore – easy distribution of a Horizon Time face to other users
- Siblings: **Pebble Round 2** (\$199, round color e-paper, ships September) · **Pebble 2 Duo** (\$149, B/W, waitlist)

2 • Watchy by SQFMI **BEST HACKABLE**

~\$50 kit + injection-molded case · [Crowd Supply](#), [Mouser](#), SparkFun

- 1.54" 200×200 **monochrome e-paper** · ESP32-S3 · WiFi + BLE · accelerometer · RTC
- Fully open hardware and firmware – program in Arduino/C++, MicroPython, or ESP-IDF
- WiFi lets the watch fetch sunrise/sunset or sync with the Paragonday backend directly – no phone app needed
- A Horizon Time face is a single C++ class drawing to a 200×200 canvas

3 • LILYGO T-Wrist **CHEAPEST PROTOTYPE**

\$13-26 bare-board wearable · lilygo.cc

- Same 1.54" 200×200 e-paper + ESP32 · optional GPS
- Community firmware [qpaperOS](#) + free 3D-printed case designs
- Ideal for testing the Horizon Time rendering cheaply before committing to nicer hardware

Circular Options

Round e-ink is rare – the **Pebble Round 2 is effectively the only circular programmable e-paper watch** on the market.

Pebble Round 2

ONLY ROUND PICK

\$199 · pre-order, ships September 2026 at repebble.com

- 1.3" **round 64-color e-paper touchscreen**, 260×260 at 283 DPI, much thinner bezel than the 2016 original
- 41.5 mm × 8.1 mm · 316 stainless steel · Matte Black, Brushed Silver, Polished Rose Gold
- 10-14 day battery · 3 ATM · accelerometer, compass, dual mics, LED backlight
- Same fully open-source PebbleOS + SDK as the Time 2 – a Horizon Time face ports directly
- A round dial is arguably the *best* canvas for Horizon Time: the sun arc maps naturally onto a circular face

Hackability: yes, first-class – no jailbreak needed

Custom watchfaces are the *intended* use, not a hack. The Round 2 is platform **gabbro** in the official Pebble SDK: write faces in C or pure JavaScript, test in the emulator (`pebble install emulator --gabbro`) or in the browser via the revived [CloudPebble](https://cloudpebble.com), then sideload or publish to the Pebble Appstore. Sunrise/sunset/location data flows in via the PebbleKit JS companion on the phone. The community has already shipped 2,120+ apps and faces for the Time 2 and Round 2 – and a Horizon Time face can be built and demoed in the emulator before hardware ships in September.

DIY round e-ink is a dead end for now: round e-paper modules are essentially not sold (Waveshare's round modules are LCD), so there is no Watchy-style circular build path.

Fit with Horizon Time

- **E-ink is ideal:** always-on, refreshes once a minute, and the sun-arc changes slowly through the day – a perfect match for partial-refresh e-paper.
- **Color, polish, distribution** → Pebble Time 2. The watchface SDK supports canvas drawing; the phone app feeds location and sunrise data.
- **Standalone, hackable, cheap** → Watchy. Direct WiFi to sunrise APIs or Paragonday's backend; full control of every pixel and the deep-sleep cycle.
- **Suggested path:** prototype the face rendering once at 200×200 mono (Watchy), then port the same arc math to a PebbleOS face.

Considered and Excluded

Bangle.js 2 and PineTime use always-on transfective / memory-in-pixel LCDs, not e-paper. Garmin Instinct is MIP with a more closed Connect IQ SDK.

Sources

repebble.com/watch · ericmigi.com – PebbleOS open source · linuxiac.com – Pebble is back · watchy.sqfmi.com · github.com/sqfmi/Watchy · Crowd Supply – Watchy · LILYGO T-Wrist · qpaperOS